

SeokHyeon (Lucas) Kong

CONTACT INFORMATION	Moore 1st floor University of Pennsylvania 200 South 33rd Street, Philadelphia, PA 19104	Personal Website: lucasshkong.github.io Email: lucasshkong@gmail.com
RESEARCH INTERESTS	My research interests lie in <i>Quantum Computer Architecture</i> , currently focusing on Quantum Error Correction.	
EDUCATION	University of Pennsylvania , Philadelphia, PA	Aug 2026
	• Ph.D. in Computer and Information Science • Advisor: Prof. Gushu Li	
	Sungkyunkwan University (SKKU) , Suwon, Korea	Mar 2020 – Feb 2026
	• B.S. in Electronic and Electrical Engineering • B.S. in Advanced Semiconductor Engineering – System Architecture Track • <i>Graduated with Summa Cum Laude</i> (GPA: 4.36/4.5)	
	The Pennsylvania State University , University Park, PA	Fall 2024
	• Exchange Student in Computer Engineering • <i>Dean's List</i> (Grade: 4.0/4.0)	
PUBLICATIONS	[1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy SeokHyeon Kong, Donghwan Kim, Euiseong Seo, and Kiwan Maeng CAL 2025 – IEEE Computer Architecture Letters [2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system SeokHyeon Kong and Hayoung Oh JKIICE 2023 – Journal of the Korean Institute of Information and Communication Engineering	
POSTER PRESENTATION	[3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl Qcenter – Quantum Information Research Support Center, Korea	
EXPERIENCE	Hanwha Systems , Seongnam, Korea	Jan 2026 – Jul 2026
	<i>Software Engineer in the Avionics System R&D Team</i> • Collaborated with The Boeing Company on ELAD (Eagle Large Area Display) for F-15 aircraft.	
	IonQ , Remote	Aug 2025 – Sep 2025
	<i>Research Mentee with Dr. Denny Dahl</i> • Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3]. • Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].	

